Death Avoidance Accounts for 90% of the Routing Signal in LLM Agent Tool Use

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Abstract

Modular LLM agents that route between neural reasoning and deterministic computation promise improved reliability in rule-governed environments, yet understanding *what* drives effective routing remains unclear. Per-step oracle labeling degenerates in sparse-reward settings—in ScienceWorld, 97.4% of steps yield tied outcomes, rendering step-level analysis uninformative. We propose episode-level counterfactual analysis: running complete episodes under pure strategies and comparing final outcomes to derive per-episode routing labels. Across 245 episodes with a Qwen2.5-7B-Instruct backbone, death avoidance accounts for approximately 90% of the routing signal (Death Avoidance Factor), with near-perfect cross-task correlation ($r = 0.97$). These results indicate that deterministic computation's value is concentrated in catastrophic failure prevention rather than positive task guidance.

1 Introduction

When a Qwen2.5-7B-Instruct agent attempts to boil water in ScienceWorld (Wang et al., 2022), it must track temperature thresholds, container states, and heat source interactions across dozens of steps. Internal LLM reasoning hallucinates these state transitions—predicting water has boiled when the stove was never activated—triggering agent death with a -100 penalty. Deterministic external computation (environment simulators, code interpreters) eliminates such hallucinations by executing exact state transitions. The natural question: when should an agent route computation externally versus reasoning internally?

Existing routing approaches (Schick et al., 2023; Paranjape et al., 2023; Lu et al., 2023) frame tool selection as a capability or efficiency decision—matching subtasks to tools that can solve them. This framing ignores a potentially dominant signal: routing to *avoid catastrophic failure*. Understanding what routing actually learns requires counterfactual analysis, but the standard per-step oracle approach degenerates under sparse rewards. On ScienceWorld, where non-zero reward arrives only at episode termination, 97.4% of steps produce tied intermediate scores under both strategies across 245 episodes—no step-level signal exists to train a router (§2.1).

We propose *episode-level counterfactual analysis*: run complete episodes under pure-internal (ReAct (Yao et al., 2023)) and pure-deterministic strategies, then compare episode outcomes to generate routing labels (§2.1). Across 11 ScienceWorld task types (245 episodes), the full deterministic advantage is $\Delta_{\text{full}} = +29.60$ points. Excluding death episodes, this gap collapses to $\Delta_{\text{nd}} = +2.84$—a Death Avoidance Factor (DAF; §2.3) of 90.41%. The per-task correlation between death rate differential and performance gap is $r = 0.9698$ ($p = 8.12 \times 10^{-7}$), confirming death avoidance as the primary mechanism (§3.2). Death rates differ sharply: 42.9% for internal reasoning versus 16.3% for deterministic computation. Of 245 episodes, 86 prefer deterministic, 9 prefer internal, and 150 show no preference—96.3% of oracle decisions match always-deterministic (§3.3).

Our contributions:

1. **Episode-level counterfactual methodology.** We introduce episode-level counterfactual analysis with death-aware labeling and the DAF decomposition metric, overcoming per-step oracle degeneracy in sparse-reward environments (§2.1, §2.3).

2. **Death avoidance dominates routing signal.** Across 11 task types (245 episodes), death avoidance accounts for 90.41% of the routing signal ($\Delta_{\text{nd}} = +2.84$, 9.6% of Δ_{full}). Router training as a diagnostic probe confirms learnable routing structure at the representation

level (§3.4).

2 Method

2.1 Counterfactual Episode Collection

Per-step oracle labeling assigns routing labels by comparing single-step outcomes under each modality. Under sparse terminal rewards, this degenerates: on ScienceWorld (Wang et al., 2022), 97.4% of steps produce identical intermediate scores regardless of modality, yielding no routing signal. We instead perform episode-level counterfactual analysis.

For each task instance, we execute two complete episodes under identical initial conditions. The *pure-internal* episode follows the ReAct paradigm (Yao et al., 2023), where the LLM interleaves chain-of-thought reasoning with environment actions. The *pure-deterministic* episode uses the environment simulator to compute exact state transitions. In both cases the LLM selects which action to take; deterministic computation replaces only the state-transition prediction—equivalent to giving the agent a perfect world model.

Each episode pair produces a tuple $(s^{\text{int}}, s^{\text{det}}, d^{\text{int}}, d^{\text{det}})$, where s denotes the terminal score and $d \in \{0, 1\}$ indicates agent death. Episodes are capped at 30 steps. The LLM backbone is Qwen2.5-7B-Instruct served via vLLM. ScienceWorld spans 30 task types with a -100 death penalty; after prescreening, we collect 245 episode pairs across 11 task types (20–40 per type).

2.2 Death-Aware Labeling

Each counterfactual pair receives an episode-level routing label based on $(s^{\text{int}}, s^{\text{det}}, d^{\text{int}}, d^{\text{det}})$. Scores are integer-valued, so we use exact (in)equality. The label is assigned by four mutually exclusive cases:

1. **Death asymmetry**: $d^{\text{int}} \neq d^{\text{det}} \rightarrow$ surviving strategy preferred.
2. **Both dead**: $d^{\text{int}} = d^{\text{det}} = 1 \rightarrow$ `no_preference`.
3. **Score gap**: both survive, $s^{\text{det}} \neq s^{\text{int}} \rightarrow$ higher-scoring strategy preferred.
4. **Tied alive**: both survive, $s^{\text{det}} = s^{\text{int}} \rightarrow$ `no_preference`.

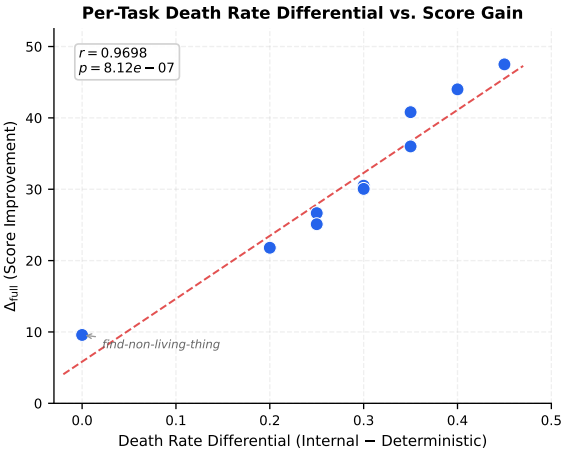


Figure 1: Per-task death rate differential (internal minus deterministic) vs. Δ_{full}. Each point represents one of 11 task types. Regression: $r = 0.9698$ ($p = 8.12 \times 10^{-7}$).

2.3 Death Avoidance Factor

To quantify the contribution of death avoidance to routing signal, we define the Death Avoidance Factor (DAF):

$$\text{DAF} = 1 - \frac{\Delta_{\text{nd}}}{\Delta_{\text{full}}} \quad (1)$$

where Δ_{full} is the mean score gap $(s^{\text{det}} - s^{\text{int}})$ across all episodes and Δ_{nd} is the same gap restricted to episodes where neither strategy results in death. DAF $= 1$ indicates all signal stems from death avoidance; DAF $= 0$ indicates none does. We report results in §3.2.

3 Experiments

3.1 Setup

We evaluate on ScienceWorld (Wang et al., 2022), a text-based interactive environment for elementary science tasks with sparse terminal rewards and a -100 death penalty. Using the methodology in §2.1, we collect 245 episode pairs across 11 task types (20–40 per type). The LLM backbone is Qwen2.5-7B-Instruct served via vLLM with episodes capped at 30 steps.

3.2 Death Avoidance Decomposition

The full performance gap between deterministic and internal strategies across 245 episodes is $\Delta_{\text{full}} = +29.60$ points. Excluding all episodes where either strategy results in death, the gap collapses to $\Delta_{\text{nd}} = +2.84$. The resulting DAF (§2.3)

Strategy	Score	95% CI	Death %
Always-Internal	−39.07	[−46.40, −32.27]	42.9
Always-Det.	−9.47	[−14.71, −4.29]	16.3
Random	−22.38	[−28.67, −15.75]	27.8
Oracle	−6.58	[−11.54, −1.70]	14.3
Trained Router	−10.70	[−16.33, −5.27]	17.6

Table 1: Episode-weighted mean scores on ScienceWorld (11 task types, 245 episodes). 95% Bootstrap CIs. The oracle selects the higher-scoring strategy per episode; 96.3% of its decisions match always-deterministic.

is 90.41%: approximately 90% of the deterministic strategy's advantage is attributable to death avoidance.

This concentration is consistent across task types. The Pearson correlation between per-task death rate differential and Δ_{full} is $r = 0.9698$ ($p = 8.12 \times 10^{-7}$; Figure 1). The internal strategy dies in 42.9% of episodes versus 16.3% for deterministic—a 2.6× ratio driving the bulk of the score difference.

The label distribution reflects this asymmetry: 86 `det_preferred`, 9 `internal_preferred`, 150 `no_preference`. When internal reasoning does not cause death, it performs comparably to deterministic computation.

One task reveals residual non-death signal: `find-non-living-thing` ($n = 40$) has zero deaths yet $\Delta_{\text{full}} = +9.58$, entirely from non-death differences. Among the 10 death-prone task types, however, death avoidance overwhelms non-death signal.

3.3 Baselines

Table 1 reports episode-weighted mean scores and death rates. Always-deterministic (−9.47) outperforms always-internal (−39.07) by 29.60 points, consistent with Δ_{full}. The oracle improves only modestly to −6.58, with 96.3% of its decisions matching always-deterministic—*oracle degeneracy* indicating the optimal routing policy closely resembles unconditional deterministic computation. Random routing (−22.38) falls between the two pure strategies.

Keyword-based routing fails structurally: ScienceWorld actions use generic verbs (`pick up`, `move`, `connect`) rather than science-specific terms, causing match rates to cluster at extremes—near-total (85.8%, degenerating to always-deterministic) or negligible (<5%, degenerating to always-internal).

The trained router (§3.4) scores −10.70, 1.23 points below always-deterministic, with comparable death rate (17.6% vs. 16.3%).

3.4 Router Training as Diagnostic

We investigate whether episode-level counterfactual labels can train a step-level router, as a diagnostic probe rather than a routing system proposal.

Classification head. A linear head appended to the LLM's hidden representations degenerates to always-deterministic (0% internal recall) across all configurations (class weighting, focal loss, undersampling, temperature scaling).

Implicit router. We fine-tune Qwen3-1.7B to internalize routing decisions. Training configuration significantly affects results. With gradient accumulation GA=8 (effective batch size 32), the model achieves generation accuracy of 65.4% with internal recall of 51.5%—the only non-degenerate configuration. With GA=1 (effective batch size 4), logit-level accuracy reaches 74.5% (balanced accuracy 79%), but generation degenerates to 53.3% accuracy with only 26.7% internal recall—a 21.2 percentage point logit–generation gap. Effective batch size determines whether learned patterns transfer to generation.

Across four seeds (GA=8), balanced accuracy is $58.1 \pm 0.8\%$—8 points above chance. Internal recall varies from 51.5% to 81.8%, indicating seed-dependent boundaries with consistent weak discriminability. End-to-end (Table 1): the router's 97.0% simulator invocation ratio means it routes internally only 3% of the time—weak signal with unstable thresholds, consistent with minimal routing structure beyond death avoidance.

4 Related Work

Tool-Use Learning. Toolformer (Schick et al., 2023) inserts API calls via self-supervised filtering; ART (Paranjape et al., 2023) delegates tool selection to LLM reasoning chains; Chameleon (Lu et al., 2023) composes modules under LLM-as-planner orchestration; ReTool (Feng et al., 2025) applies RL for implicit code-interpreter invocation. These methods learn *which tool* to use but do not analyze what signal drives routing. We generate labels via *episode-level counterfactual analysis* and diagnose what the signal encodes—revealing ∼90% is death avoidance.

Adaptive Computation. CogRouter (Yang et al., 2026b) achieves 82.3% on ScienceWorld (success rate, different evaluation protocol) via cognitive depth routing. ARES (Yang et al., 2026a) reduces token cost by 52.7% with per-step reasoning effort selection. Mixture-of-Recursions (Bae et al., 2025) applies adaptive depth via per-token recursive routing. These route *how much* to reason; we route *which modality*. None quantify routing signal composition.

World Model Routing. ITP (Liu et al., 2026) augments agents with LLM-based world model lookahead; Dyna-Think (Yu et al., 2025) integrates world model simulation into reasoning, matching 685B performance with 32B at half the token cost. Both route to *approximate* world models inheriting LLM hallucinations. Our deterministic modality provides *exact* state transitions; counterfactual analysis shows this exactness matters almost exclusively for death avoidance.

Agent Frameworks. ReAct (Yao et al., 2023) interleaves reasoning and acting and serves as our internal-reasoning baseline. MTRouter (Zhang et al., 2026) routes between LLMs for cost optimization on ScienceWorld. Our work asks what a learned router captures—the answer is dominated by catastrophic-failure prevention, suggesting safety-aware computation allocation over general reasoning enhancement.

5 Discussion

Cross-environment evidence. The death-avoidance finding in ScienceWorld raises a natural question: does routing signal persist in environments without catastrophic failure? A preliminary 50-problem pilot on MATH Level 4–5, using code interpreter as the deterministic modality, suggests not. Internal reasoning achieved 80% accuracy versus 72% for code interpreter, with 88% of problems showing no modality preference (label distribution: 1 deterministic-preferred, 5 internal-preferred, 44 no-preference). In the absence of death risk, the routing signal vanishes or reverses—consistent with death avoidance as the primary driver of the deterministic strategy's advantage observed in ScienceWorld. A larger-scale replication with improved code execution (addressing sandbox limitations in the pilot) is in progress.

Scope and future directions. Our analysis is grounded in a single interactive environment with a specific failure mode (the -100 death penalty). Extending the counterfactual methodology to environments with graded failure costs, embodied agents with continuous action spaces, or multi-turn dialogue systems would clarify how broadly death avoidance—or analogous catastrophic-failure avoidance—accounts for apparent routing signal. Step-level counterfactual labeling, rather than episode-level, is a complementary direction that could reveal finer-grained routing structure masked by our current aggregation.

6 Conclusion

We presented episode-level counterfactual analysis for diagnosing routing behavior in modular LLM agents. Applied to 245 episodes across 11 ScienceWorld task types, death avoidance accounts for 90.41% of the routing signal (Δ_{full} = +29.60 vs. Δ_{nd} = +2.84), corroborated by near-perfect cross-task correlation (r = 0.9698), oracle degeneracy (96.3% agreement with always-deterministic), and near-random trained-router accuracy (58.1%). Preliminary cross-environment evidence (§5) suggests this finding reflects a structural property of catastrophic-failure environments rather than a ScienceWorld artifact.

Limitations

Our analysis uses a single benchmark (ScienceWorld) with a single LLM backbone (Qwen2.5-7B-Instruct), and generalization to other interactive environments and model scales remains to be established. The -100 death penalty may amplify DAF; environments with less catastrophic penalties could exhibit lower DAF, though the qualitative finding may still hold. Step-level labeling with denser rewards is a natural extension. The logit–generation gap in the implicit router warrants further investigation.

Labeling quality. A potential concern is oracle circularity: whether episode-level labels reflect tie-breaking artifacts rather than genuine modality differences. We audit all 245 labels. Of the 150 `no_preference` labels (61.2%), 115 (46.9%) are both-alive ties and 35 (14.3%) are both-dead—all excluded from router training. The remaining 95 training labels are entirely genuine: 75 from death asymmetry (70 deterministic-preferred, 5 internal-preferred) and 20 from score comparison

with both strategies surviving (16 deterministic-preferred, 4 internal-preferred). Zero labels depend on tie-breaking. Retraining the router using only death-asymmetry labels yields balanced accuracy of 57.3%, within 1σ of the full-label retrain ($57.6 \pm 0.5\%$), confirming that score-comparison labels contribute negligible additional signal. The 97.4% step-level tie rate (§2.1) further motivates episode-level over step-level labeling.

Ethics Statement

This work uses ScienceWorld (Wang et al., 2022), a publicly available text-based simulation environment, and does not involve human subjects or private data. All language models are accessed via standard APIs or publicly released weights. We report computational costs: approximately 40 GPU-hours on two RTX 5090 GPUs for data collection, router training, and evaluation.

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